

Simulating Differential Equations with Euler's Method
NEUROSCI 366, Brain Function Through the Lens of Computation
Discussed: Friday, April 4, 2025

Differential equations are extremely useful for modeling the dynamics of neural systems. In this document and associated code, we will illustrate how to use MATLAB for simulating differential equations with Euler's method.

1 Exponential equations with time-varying input

Suppose that the system evolves according to

$$\tau \frac{dx}{dt} = -x + I(t), \quad (1)$$

where $\tau = 10$ ms,

$$I(t) = \begin{cases} 0 & , 0 \text{ ms} \leq t < 50 \text{ ms} \\ 2 & , 50 \text{ ms} \leq t < 100 \text{ ms} \\ 1 & , 100 \text{ ms} \leq t < 150 \text{ ms} , \\ \cos(\omega(t - t_0)) & , 150 \text{ ms} \leq t < 200 \text{ ms} \\ 0 & , 200 \text{ ms} \leq t < 250 \text{ ms} \end{cases} \quad (2)$$

$\omega = 220$ radians per second, and $t_0 = 150$ ms. The provided code implements Euler's method to simulate this differential equation in MATLAB for $0 \text{ ms} \leq t \leq 250 \text{ ms}$ and $x(0) = 0$.

NOTE: It was important to choose a small enough Δt to get a meaningful result. You may want to explore how the result and run time change as you vary the value of Δt .

2 Higher-order equations and coupled first-order ones

Now suppose that x and y evolve according to the coupled differential equations

$$\frac{dx}{dt} = y, \quad (3)$$

$$\frac{dy}{dt} = -x. \quad (4)$$

We explained in the lecture notes that this is equivalent to the second-order differential equation that generates sinusoidal oscillations. The provided code implement Euler's method to simulate these differential equations in MATLAB for $0 \leq t \leq 10$, $x_1(0) = 0$, and $x_2(0) = 1$.